

AI INNOVATORS PATHWAY

AI Entrepreneurship and Innovation

Student Workbook

6

Project-Based Lessons

9-12

Grade Range

1

Final Innovation Pitch

Name: _____ Team: _____ Class: _____

Pilot Edition | Grades 9-12

Future Ready AI

Learn. Create. Lead.

Your Innovation Challenge

Your team will investigate a real school, community, business, or career-related problem; decide whether AI can responsibly support a solution; create a concept prototype; and pitch a realistic next step.

Your responsibility: AI can support ideas and tasks, but people remain responsible for evidence, design choices, safeguards, decisions, and consequences.

What you will produce

- Problem statement and Worth-Solving Scorecard
- Customer and stakeholder discovery evidence
- AI Fit Test, role map, and concept storyboard
- Value proposition and simple business-model canvas
- Responsible AI Risk Review
- Prototype, pitch deck, presentation, and reflection

Pre-Assessment

Complete independently before beginning the module. This is a baseline, not a judgment of prior access to AI.

1. A team starts by choosing a chatbot and then searches for a problem. What should it do first?

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- B. Identify and investigate a specific human need
- C. Create a logo
- D. Estimate advertising revenue

2. Which discovery question is most likely to produce useful evidence?

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- B. Do you like AI?
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3. A stakeholder is best described as:

- A. Only the person who pays
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4. When is AI most appropriate?

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6. A strong value proposition identifies:

- A. Only the product name
- B. The people, their need, the benefit, and how the solution differs from current options
- C. Every possible feature
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7. A simple business model explains:

- A. Only how much money investors receive
- B. How value is created, delivered, resourced, and sustained
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- D. Why no partners are needed

8. Which practice best protects privacy?

- A. Collect all available data in case it becomes useful
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9. Bias risk can occur when:

- A. Data or design choices underrepresent people or reproduce unequal patterns
- B. A system processes information quickly
- C. A prototype uses paper
- D. A team receives feedback

10. Which is an access consideration?

- A. Whether users have the language, device, connectivity, ability, and support needed to use the solution
- B. Whether the logo is memorable
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12. If the highest-risk feature cannot be adequately safeguarded, the team should:

- A. Launch quickly
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13. Describe one appropriate role AI could play in a solution and one limitation that must be managed.

14. Explain what human accountability should look like in an AI-supported solution.

Lesson 1: What Makes a Problem Worth Solving?

Opportunity scan: List at least five problems you notice in school, community, work, or daily life. Do not name a solution yet.

Worth-Solving Scorecard

Candidate problem	Evidence (1-4)	Impact (1-4)	Frequency (1-4)	Access to people (1-4)	Feasible scope (1-4)	Total

Selected problem: Who experiences it, when/where does it happen, and what impact does it create?

What do we know from evidence? What are we assuming? What must we learn next?

Problem statement: [People] experience [specific problem] in [context], resulting in [impact], as shown by [evidence or learning need].

Lesson 2: Customer and Community Discovery

Stakeholder map

Group	Relationship to the problem	What we need to learn	Safe way to learn

Discovery question builder

- Tell me about the last time this problem occurred.
- What do you do now? What works and what is difficult?
- What outcome matters most to you?
- What would make a new approach difficult to use or trust?
- Who else is affected or should be involved?

Write six neutral questions. Avoid asking people to approve your idea.

Privacy rule: Record themes, not identities. Do not collect names, contact details, health information, grades, immigration status, financial details, or other sensitive information.

Discovery notes

Source type (no names)	What happens now	Needs or goals	Barriers/constraints	Surprise or contradiction

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Evidence-to-insight summary: What patterns appeared? What contradicted your assumptions? Whose perspective is still missing?

Design criteria: Our solution must... / must not...

Lesson 3: How Can AI Support the Solution?

AI Fit Test

Question	Yes / No / Unsure	Evidence or reasoning
Is there a specific task that AI can perform or support?		
Does the task involve patterns, prediction, recommendation, classification, or generation?		
Is relevant, lawful, and appropriate information available?		
Would ordinary software, a checklist, search, or a human service work better?		
Can important outputs be verified before action?		
Can a person meaningfully review, override, and explain the result?		
Is the likely benefit proportionate to the risk and resources?		

Decision: AI is / is not appropriate because...

AI Role Map

Input	AI-supported task	Output	Verification	Human decision/action

What must the system never decide or do alone? What is the non-AI fallback?

Concept storyboard

1. Before	2. Need	3. Interaction	4. AI support	5. Human review	6. Outcome

Lesson 4: Value Proposition and Business Model

Value proposition draft: For [specific people] who [need/problem], our [solution] helps them [benefit] by [how], unlike [current alternative].

Clear, Credible, Compelling Test

Test	Peer feedback	Revision
Clear: Can a listener explain it in their own words?		
Credible: Is it supported by discovery evidence?		
Compelling: Does the benefit matter to the intended people?		

Simple Business-Model Canvas

Canvas area	Team response
Problem	What verified problem are we addressing?
Customers, users, and beneficiaries	Who uses, decides, pays, benefits, or may be affected?
Value proposition	What meaningful outcome will improve?
Solution and AI role	How does it work, and what limited task does AI support?
Channels and engagement	How will people learn about, access, and use it?
Key partners and resources	What people, expertise, data, technology, permissions, or relationships are required?

Canvas area	Team response
Costs and resource needs	What must be funded, built, maintained, or supported?
Sustainability	Revenue, sponsorship, school/community support, service model, or other way the work continues.
Success measures	How will we know it is useful, fair, safe, accessible, and sustainable?

Riskiest assumption and smallest low-cost test:

Lesson 5: Responsible Innovation Review

Rate each risk: likelihood 1-4 and severity 1-4. Risk priority = likelihood x severity. A low number does not eliminate responsibility.

Risk area	Guiding question	L	S	Priority	Safeguard / owner
Purpose and necessity	Is the AI feature necessary and proportionate?				
Privacy and data	What is collected, inferred, stored, shared, or retained? What can be minimized?				
Bias and fairness	Who may be underrepresented, misclassified, stereotyped, or treated unequally?				
Access and inclusion	What language, disability, device, connectivity, cost, or support barriers exist?				
Accuracy and transparency	How will people know AI is involved, verify output, and understand limits?				
Harm and misuse	How could this fail, mislead, exclude, manipulate, or be used for another purpose?				
Human accountability	Who reviews, decides, overrides, responds, and documents?				
Appeal and remedy	How can a person question a result, get help, and correct harm?				
Monitoring and stop rules	What will be checked, and what would trigger a pause or removal?				

Highest remaining risk: Who could be affected, what evidence is missing, and what must happen before further testing?

Revised concept: What did your team limit, add, remove, disclose, or place under human control?

Red-Team Feedback

How might this fail, exclude, mislead, or be misused?

Our response and revision:

Lesson 6: Build and Pitch the Solution

Prototype planner

Prototype type	What it should demonstrate	Materials/tools	Who will test it	What we will observe

- Paper screens or interface sketch
- Storyboard or service journey
- Clickable slide mockup
- Workflow demonstration
- Role-play or service blueprint
- Physical model or annotated concept poster

Test notes: One strength, one confusion, one question, and one revision:

Eight-Slide Pitch Planner

Slide	Purpose	Our key message / evidence
1	Solution and team	Name, one-sentence promise, team
2	Problem worth solving	People, context, impact, evidence
3	Discovery	What stakeholders do now, needs, constraints, insight
4	Solution and AI role	Experience, AI-supported task, alternative considered
5	Value proposition	Benefit, differentiation, reason to trust
6	Business/community model	Delivery, partners/resources, costs, sustainability, success
7	Responsible innovation	Top risks, safeguards, human owner, appeal/fallback
8	Prototype and next test	Demonstration, feedback, revision, specific ask

Slide	Purpose	Our key message / evidence

Questions we expect and our evidence-based responses:

Final Challenge Submission Checklist

- Our problem statement is specific and supported by evidence.
- We included multiple perspectives and did not collect sensitive personal information.
- We can explain why AI is appropriate—or why we limited its role.
- We mapped inputs, outputs, verification, human authority, and fallback.
- Our value proposition and business/community model are realistic.
- Our risk review changed the design and names accountable people.
- Our prototype demonstrates the experience or workflow.
- Our pitch is 3-5 minutes, evidence-based, accessible, and candid about limitations.
- Every team member can explain the concept and answer questions.

Peer Feedback

Strong evidence or design decision:

Question or uncertainty:

Responsible next-test recommendation:

Performance Rubric

Criterion	4	3	2	1
Problem and evidence	Specific, consequential, scoped, multiple evidence points	Clear; some evidence	Broad or assumption-led	Missing/disconnected
Discovery	Multiple perspectives shape design	Relevant perspectives included	Limited or weakly connected	Little/no evidence
AI fit and oversight	Precise role, alternatives, limits, verification, fallback, authority	Mostly clear	Vague/unnecessary or weak oversight	AI slogan or no accountability
Value and model	Specific value; coherent delivery, resources, sustainability, success	Plausible with gaps	Generic/incomplete	Missing

Criterion	4	3	2	1
Responsible innovation	Risks evaluated; safeguards change design	Major risks addressed	General safeguards	Risks ignored
Prototype	Clear experience/workflow; tested and revised	Communicates concept	Partial/confusing	Missing
Pitch	Clear, evidence-based, paced, candid; strong Q&A	Communicates core idea	Claims outweigh evidence	Incomplete
Collaboration/reflection	Roles, contribution, feedback, next steps documented	Most contribute	Uneven/superficial	Missing

Post-Assessment

Complete independently after the pitch. Use evidence and responsible AI principles.

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Individual Reflection

1. What discovery evidence most changed your team's thinking? Why?

2. Where did AI add genuine value? Where did your team limit or reject AI?

3. Which risk was hardest to address? What human accountability is still required?

4. What feedback improved the prototype or pitch?

5. What was your specific contribution, and what would you do differently next time?

6. Which career skills did you practice? Name at least three and give one example.

7. What is the smallest responsible next test for this concept?

Quick Glossary

Term	Plain-language meaning
Artificial intelligence (AI)	Computer systems that perform tasks associated with learning from data, recognizing patterns, predicting, recommending, classifying, or generating content.
Bias	A systematic pattern that can produce unfair or distorted outcomes.
Customer	A person or organization that chooses or pays for a product or service; not always the same as the user or beneficiary.
Data minimization	Collecting and retaining only the information necessary for a clearly defined purpose.
Human oversight	A meaningful process in which a trained person has information and authority to review, change, pause, or override an AI-supported result.
Prototype	A low-cost representation used to communicate and test an idea before full development.
Stakeholder	A person or group that may affect or be affected by a decision or solution.
Value proposition	A clear explanation of who benefits, what need is addressed, what outcome improves, and why the approach is useful.